Campus Event Financial Planning – Complete Python Solution

Problem Statement:

Develop a Python program to calculate the total expenditure of a campus event using venue, equipment, catering, transport, and certificate costs. Determine participant contributions, calculate sponsorship shortfalls, and find the minimum number of registrations required to break even.

The solution uses only your assigned Python syllabus concepts.

1. Computational Thinking

Input:

- Venue cost.
- Equipment cost.
- Catering cost.
- Transport cost.
- Certificate cost.
- Sponsorship amount.
- Registration fee per participant.

Processing:

1. Read all event expenses.
2. Calculate total event expenditure.
3. Read the sponsorship amount.
4. Calculate the remaining amount after sponsorship.
5. Read registration fee per participant.
6. Calculate the number of participants required to cover the remaining amount.
7. Calculate expected registration contributions.
8. Determine whether the event reaches break-even.
9. Display the financial report.

Output:

- Individual expenses.
- Total expenditure.
- Sponsorship amount.
- Sponsorship shortfall.
- Registration fee.
- Minimum registrations required.
- Expected registration contribution.
- Break-even status.

2. Financial Planning Rules

Assume the following example:

Expense	**Amount**
Venue	₹10,000
Equipment	₹5,000
Catering	₹15,000
Transport	₹4,000
Certificates	₹2,000
Total expenditure	₹36,000

Additional financial details:

Particular	**Amount**
Sponsorship	₹12,000
Registration fee per participant	₹300
Expected participants	100

Important formulas:

Total expenditure:

$$ T=V+E+C+Tr+Ce $$

Remaining amount after sponsorship:

$$ R=T-S $$

Minimum registrations required:

$$ N=\left\lceil\frac{R}{F}\right\rceil $$

Where:

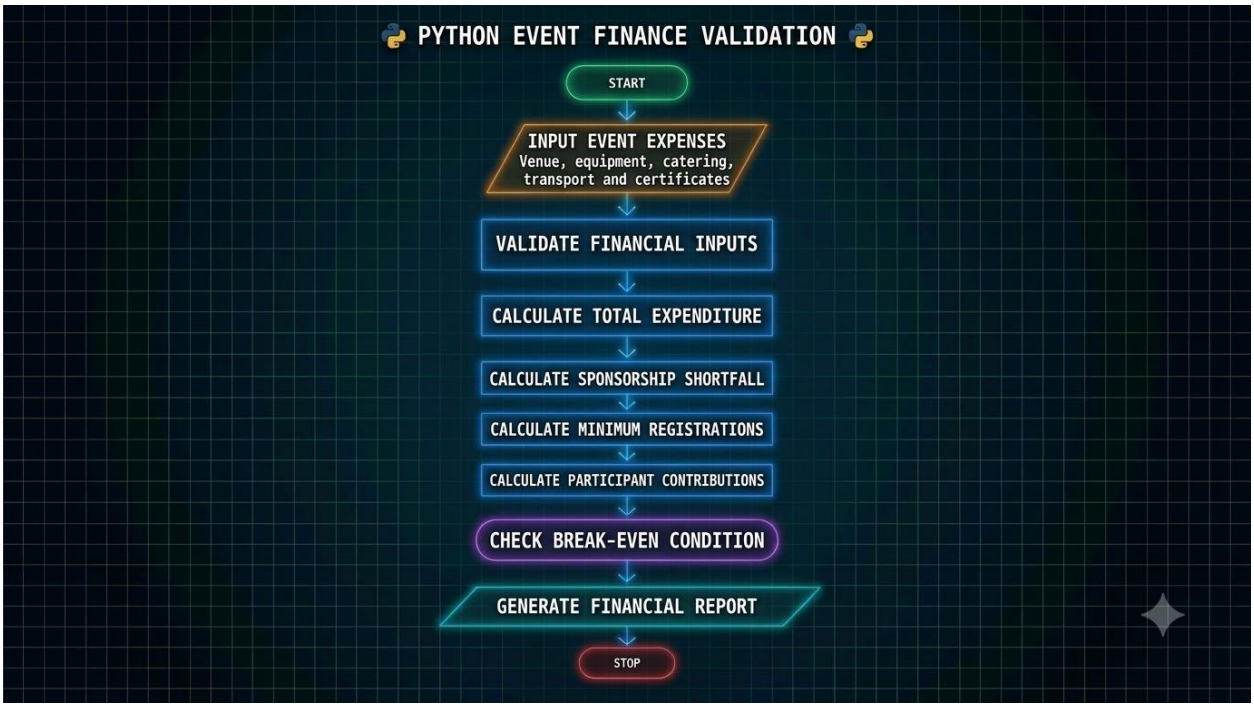
- T = Total expenditure
- S = Sponsorship
- F = Registration fee
- N = Minimum registrations

3. Algorithm

1. Start.
2. Input venue cost.
3. Input equipment cost.
4. Input catering cost.
5. Input transport cost.
6. Input certificate cost.
7. Input sponsorship amount.
8. Input registration fee.
9. Input expected participants.
10. Validate all financial inputs.
11. Calculate total expenditure.
12. Calculate remaining expenditure after sponsorship.
13. Calculate minimum registrations required.
14. Calculate total participant contributions.
15. Calculate total available funds.

16. Compare available funds with total expenditure.
17. Display break-even status.
18. Display the complete financial report.
19. Stop.

4. Flowchart



5. Pseudocode

START

INPUT venue_cost

INPUT equipment_cost

INPUT catering_cost

INPUT transport_cost

INPUT certificate_cost

INPUT sponsorship

```
INPUT registration_fee
INPUT expected_participants

IF any input is negative:
   DISPLAY "Invalid Financial Details"

ELSE IF registration_fee <= 0:
   DISPLAY "Invalid Registration Fee"

ELSE:

   total_expenses = venue_cost + equipment_cost
                  + catering_cost + transport_cost
                  + certificate_cost

   remaining_amount = total_expenses - sponsorship

   IF remaining_amount <= 0:
      minimum_registrations = 0

   ELSE:
      minimum_registrations =
         remaining_amount / registration_fee
         rounded up to next integer

   participant_contribution =
```

```
    expected_participants * registration_fee

  total_funds = sponsorship + participant_contribution

  IF total_funds >= total_expenses:
    status = "BREAK EVEN ACHIEVED"

  ELSE:
    status = "BREAK EVEN NOT ACHIEVED"

  shortfall = total_expenses - total_funds

  DISPLAY financial report

STOP
```

6. Complete Python Program

```
# Campus Event Financial Planning

# Input event expenses
venue = float(input("Enter venue cost: "))
equipment = float(input("Enter equipment cost: "))
catering = float(input("Enter catering cost: "))
transport = float(input("Enter transport cost: "))
certificates = float(input("Enter certificate cost: "))

# Input financial details
```

```
sponsorship = float(input("Enter sponsorship amount: "))

registration_fee = float(
    input("Enter registration fee per participant: ")
)

expected_participants = int(
    input("Enter expected number of participants: ")
)

# Validate financial inputs
if (venue < 0 or equipment < 0 or catering < 0 or
        transport < 0 or certificates < 0 or
        sponsorship < 0 or expected_participants < 0 or
        registration_fee <= 0):

    print("\nInvalid Financial Details!")

else:

    # Store expenses in a list
    expenses = [
        venue,
        equipment,
        catering,
        transport,
```

```
    certificates
]

# Calculate total expenditure
total_expenses = 0

for cost in expenses:
    total_expenses = total_expenses + cost

# Calculate sponsorship shortfall
remaining_amount = total_expenses - sponsorship

if remaining_amount < 0:
    remaining_amount = 0

# Calculate minimum registrations required
if remaining_amount == 0:
    minimum_registrations = 0

else:
    minimum_registrations = int(
        remaining_amount / registration_fee
    )

    if minimum_registrations * registration_fee < remaining_amount:
        minimum_registrations = minimum_registrations + 1
```

```
# Calculate participant contributions
participant_contribution = (
    expected_participants * registration_fee
)

# Calculate total available funds
total_funds = sponsorship + participant_contribution

# Calculate final financial shortfall
if total_funds < total_expenses:
    financial_shortfall = total_expenses - total_funds

else:
    financial_shortfall = 0

# Calculate surplus
if total_funds > total_expenses:
    surplus = total_funds - total_expenses

else:
    surplus = 0

# Determine break-even status
if total_funds >= total_expenses:
    status = "BREAK EVEN ACHIEVED"
```

```
else:
    status = "BREAK EVEN NOT ACHIEVED"

# Store financial details in a tuple
financial_details = (
    total_expenses,
    sponsorship,
    remaining_amount,
    registration_fee,
    expected_participants,
    participant_contribution,
    total_funds,
    minimum_registrations,
    financial_shortfall,
    surplus
)

# Tuple unpacking
(total, sponsor, remaining, fee, participants,
 contribution, funds, minimum, shortfall,
 surplus) = financial_details

# Display financial statement
print("\n========================================")
print("       CAMPUS EVENT FINANCIAL REPORT")
```

```
print("==========================================")

print("Venue Cost         : Rs.", venue)
print("Equipment Cost     : Rs.", equipment)
print("Catering Cost      : Rs.", catering)
print("Transport Cost     : Rs.", transport)
print("Certificate Cost   : Rs.", certificates)

print("------------------------------------------")

print("Total Expenditure  : Rs.", total)
print("Sponsorship        : Rs.", sponsor)
print("Sponsorship Shortfall: Rs.", remaining)

print("------------------------------------------")

print("Registration Fee   : Rs.", fee)
print("Expected Participants:", participants)

print("Participant Contribution: Rs.", contribution)

print("Minimum Registrations Required:", minimum)

print("------------------------------------------")

print("Total Available Funds: Rs.", funds)
```

```
    print("Financial Shortfall  : Rs.", shortfall)
    print("Financial Surplus    : Rs.", surplus)

    print("----------------------------------------")

    print("Break-Even Status:", status)

    print("========================================")
    print("       FINANCIAL PLANNING COMPLETED")
```

7. Sample Input

Enter venue cost: 10000

Enter equipment cost: 5000

Enter catering cost: 15000

Enter transport cost: 4000

Enter certificate cost: 2000

Enter sponsorship amount: 12000

Enter registration fee per participant: 300

Enter expected number of participants: 100

8. Sample Output

```
========================================
      CAMPUS EVENT FINANCIAL REPORT
========================================
```

Venue Cost : Rs. 10000.0

Equipment Cost : Rs. 5000.0

Catering Cost : Rs. 15000.0

Transport Cost : Rs. 4000.0

Certificate Cost : Rs. 2000.0

Total Expenditure : Rs. 36000.0

Sponsorship : Rs. 12000.0

Sponsorship Shortfall: Rs. 24000.0

Registration Fee : Rs. 300.0

Expected Participants: 100

Participant Contribution: Rs. 30000.0

Minimum Registrations Required: 80

Total Available Funds: Rs. 42000.0

Financial Shortfall : Rs. 0

Financial Surplus : Rs. 6000.0

Break-Even Status: BREAK EVEN ACHIEVED

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FINANCIAL PLANNING COMPLETED

9. Step-by-Step Calculation

Step 1: Calculate total expenditure

$$ 10000+5000+15000+4000+2000=36000 $$

Step 2: Calculate sponsorship shortfall

$$ 36000-12000=24000 $$

Step 3: Calculate minimum registrations

Each participant contributes ₹300.

$$ N=\left\lceil\frac{24000}{300}\right\rceil $$

$$ N=80 $$

Revenue

Total cost

Loss

Profit

102030405060500100015000uantity$Q*

$Q^*=\frac{F}{p-v}=\frac{\text{600}}{\text{35}-\text{15}}=\text{30}$

Q* is the break-even quantity where total revenue matches total cost.

F

$

v

$/unit

\(p\)

$/unit

Give feedback

Step 4: Calculate participant contributions

\[100\times300=30000 \]

Step 5: Calculate total available funds

\[12000+30000=42000 \]

Step 6: Calculate surplus

\[42000-36000=6000 \]

Break-even achieved

The event requires at least 80 registrations and has 100 expected participants. Therefore, the event generates a surplus of ₹6,000 under the assumed financial policy.

10. Additional Test Cases

Situation	**Expected result**
Sponsorship covers all expenses	0 registrations required
Sponsorship is zero	Participants cover full expenses
Registration fee is zero	Invalid registration fee
Registration fee is negative	Invalid financial details
Expected participants are zero	No participant contribution
Available funds equal expenses	Break-even achieved
Available funds exceed expenses	Surplus
Available funds are below expenses	Financial shortfall

11. Assigned Python Syllabus Concepts Used

Syllabus concept	Application
Computational thinking	Breaking financial planning into steps
Algorithm	Step-by-step calculations
Flowchart	Program execution flow
Pseudocode	Logical representation
Variables	Expenses, sponsorship, contributions
Data types	Float, integer, string
Operators	+, -, *, /, <, >=
Conditional statements	if, elif, else
Loops	for loop
Lists	Store expense values
Tuples	Store financial details
Tuple operations	Tuple storage and access
Tuple unpacking	Extract financial values
Strings	Financial report and status
Built-in functions	input(), print(), int()

Complete syllabus-based solution

Important: This is an illustrative campus-event budgeting model. It assumes the registration fee is fixed, all expected participants pay, and sponsorship is available to cover event expenses.